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Unit 5 : Session 6 : SLO 2 | ARM Thumb Instruction Set

I. Multiple Choice

1. b) To reduce code size and memory usage.
2. b) 16-bit.
3. c) They provide improved code density for memory-constrained systems.
4. a) By setting a bit in the CPSR (the T bit, changed via BX/BLX instructions).
5. b) Reduced program memory requirements.

II. True or False

1. TRUE.
2. FALSE – Thumb reduces memory requirements.
3. TRUE – the CPSR T bit selects the instruction set state.
4. FALSE – NEON SIMD operations need ARM/NEON state, not Thumb.
5. TRUE.

III. Scenario-Based Activity

1. Use Thumb: 16-bit instructions cut code size roughly in half versus 32-bit ARM code, so the control logic plus lookup tables fit the 64 KB flash with room for features.
2. Switching: via the CPSR T bit – executing BX/BLX with the target address bit-0 set enters Thumb state; clearing it returns to ARM state.
3. Trade-off: Thumb has a reduced instruction set – some operations need multiple Thumb instructions where one ARM instruction suffices, so performance-critical inner loops may run slower; typical practice is Thumb for bulk code, ARM state for hot paths.